XAVIER INSTITUTE OF ENGINEERING
(An Autonomous Institute under Mumbai University)
Approved by AICTE and recognized by DTE, Government of Maharashtra
Accredited with an "A+" grade by NAAC
Department of Computer Engineering
(NBA Accredited)

# Title: KPI Misinterpretation Detection System

Submitted in partial fulfilment of the requirements of

**MINI PROJECT**

**SEMESTER VI**

for the degree of

**BACHELOR OF ENGINEERING IN**

**COMPUTER ENGINEERING**

by

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UNDER THE GUIDANCE

OF

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(Assistant Professor)

**Department of Computer Engineering, Xavier Institute of Engineering, Mahim, Mumbai - 400 016 University of Mumbai (AY: 2025 - 2026)**

# Institute Vision and Mission

## Vision of Institute

To nurture the joy of excellence in a world of high technology.

## Mission of Institute

To strive to match Global Standards in technical education by interaction with industry, continuous training and development of quality of life.

# Department Vision and Mission

## Vision of Computer Department

To nurture the joy of excellence in the world of Computer Engineering.

## Mission of Computer Department

**M1:** To provide students the knowledge of computer engineering with related technical skills to solve various computer engineering problems and to have industry interaction.

**M2:** To motivate the students to acquire additional soft skills to sharpen their technical and professional skills that make them employable.

**M3:** To nurture social and professional ethics in our undergraduates and encourage them to have lifelong learning.

## Program Specific Objectives (PSO)

| | |
|---|---|
| **PSO1** | The ability to understand, analyze and apply knowledge to address computing problems. |
| **PSO2** | The ability to design and implement software solutions to meet the end users requirements. |

## Program Educational Objectives (PEOs)

| | |
|---|---|
| **PEO1** | Graduates will have the required knowledge of computer engineering, to design and develop systems that can cater to a multi-disciplinary environment. |
| **PEO2** | Graduates will be able to apply the art of self-learning to use the modern tools in solving computer engineering problems. |
| **PEO3** | Graduates will be prepared in the field of computer engineering to have lifelong-learning in the global and ethic context. |
| **PEO4** | Graduates will be able to inculcate professional and ethical attitude with good leadership qualities and have commitment to social responsibilities in their thought process. |

**Program Outcomes (PO):** A graduate of Computer engineering will have the ability:

| | |
|---|---|
| **PO1** | Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems. |
| **PO2** | Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4) |
| **PO3** | Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5) |
| **PO4** | Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8). |
| **PO5** | Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6) |
| **PO6** | The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7). |
| **PO7** | Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9) |
| **PO8** | Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams. |
| **PO9** | Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences. |
| **PO10** | Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments. |
| **PO11** | Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8) |

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# CERTIFICATE

This is to certify that the Mini Project entitled **"KPI Misinterpretation Detection System"** is a bonafide work of **Silas Shelly (01), Shannel Dcosta (14), Aidan Rebello (69)** and **Aryan Gupta (26)** submitted in partial fulfillment of the requirements of **Mini Project, Semester – VI**, for the award of the degree of "**Bachelor of Engineering"** in "**Computer Engineering"**.

**Mr. Omprakash Yadav**
**Guide**
**Assistant Professor**

Dr. Sushama Khanvilkar
H.O.D Computer Engineering

Dr. Lata Ragha
Principal

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# MINI PROJECT APPROVAL

This Mini Project entitled "**KPI Misinterpretation Detection System**" by Silas Shelly (01), Shannel Dcosta (14), Aidan Rebello (69) and Aryan Gupta (26) is approved for **MINI PROJECT SEMESTER: VI** for the degree of "**Bachelor of Engineering**" in "**Computer Engineering**".

**Examiners**:

1._____

(External Examiner)

2._____

Mr. Omprakash Yadav
(Internal Examiner)

Date:

Place: Mumbai

# Declaration

We declare that this written submission represents our ideas in our own words and where others ideas or words have been included; we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Silas Shelly (01)

Shannel Dcosta (15)

Aidan Rebello (69)

Aryan Gupta (26)

Date:

# Abstract

In the evolving landscape of data-driven decision-making, organizations rely heavily on Key Performance Indicators (KPIs) to evaluate business performance and guide strategic actions. However, traditional analytics systems often focus on isolated KPI values or basic visualizations, which can lead to misinterpretation and poor decision-making. To address these challenges, we propose "Metric Lies – KPI Misinterpretation Detection System," an intelligent analytics solution designed to analyze business KPIs and identify misleading patterns, conflicts, and anomalies. This system enhances how business analysts interpret performance metrics by providing deeper, context-aware insights using artificial intelligence.

At the core of the system lies a comprehensive data processing pipeline that supports KPI data from structured sources such as Excel files and databases. The system preprocesses the data through cleaning, normalization, and feature extraction techniques. It then analyzes key business metrics such as revenue growth, profit margin, customer churn rate, customer acquisition cost (CAC), customer lifetime value (CLV), return on investment (ROI), and cash flow. These KPIs are further categorized based on company size, including large-cap, midcap, and small-cap organizations, to provide more context specific analysis.

For generating actionable insights, the system utilizes machine learning techniques such as anomaly detection to identify unusual KPI behavior, correlation analysis to detect misleading relationships between metrics, and classification models to label KPI performance as healthy, risky, or misleading. The processed insights are then visualized using interactive dashboards and Power BI, enabling users to clearly understand KPI trends, conflicts, and potential risks. This system ultimately helps organizations make more accurate, data-driven decisions by identifying when metrics may present a false or incomplete picture of business performance.

# Acknowledgement

We the students of **Xavier Institute of Engineering, Mahim** of **Third Year Engineering** in the department of **Computer Engineering** have great pleasure in presenting our efforts of developing the project named as "**KPI Misinterpretation Detection System**".

We would like to express our heartfelt gratitude to **Dr. Sushama Khanvilkar**, **Head of the Department**, for providing us with inspiration, valuable guidance, and timely suggestions throughout our Mini Project journey.

Our immense appreciation also goes to our esteemed **Guide Mr. Omprakash Yadav** and **Coordinator Prof. Kavita Jain** for their unwavering support and the opportunity they provided us to undertake this challenging project has been truly invaluable. This experience not only allowed us to conduct extensive project work but also exposed us to a wealth of new knowledge for which we are genuinely thankful.

# CONTENTS


| | | |
|---|---|---|
| **Acknowledgment** | | **X** |
| **List of Figures** | | **XI** |
| **1** | **Introduction** | **12** |
| 1.1 | Introduction | 12 |
| 1.2 | Motivation | 13 |
| 1.3 | Problem Statement | 14 |
| 1.4 | Objectives | 15 |
| **2** | **Literature Survey** | **16** |
| 2.1 | Survey of Existing System | 16 |
| 2.2 | Limitation Existing system or research gap | 16 |
| 2.3 | Mini Project Contribution | 17 |
| **3** | **Proposed System** | **18** |
| 3.1 | Architecture/ Framework | 18 |
| 3.2 | Algorithm and Process Design | 19-20 |
| 3.3 | Details of Hardware & Software | 21 |
| 3.4 | PERT / GANTT / CPM Charts | 22-24 |
| **4** | **Results and Discussions** | **25-26** |
| **5** | **Conclusion and Future Scope** | **27** |
| **6** | **References** | **28** |

# List Of Figures

3.1. Architecture/Framework . . . . . 19
3.2. Pert Chart . . . . . .23
3.3. Gantt Chart . . . . . 24
3.4. CPM Chart . . . . . .25
4.1. KPI Sentinel dashboard . . . . . .26
4.2. KPI Predictions. . . . . 26
4.3. Cross-KPI Matrix . . . . . 27

## Lab Outcomes:

| Sr. No. | Lab Outcomes | Bloom's Taxonomy |
|---|---|---|
| On successful completion, of course, learner/student will be able to: | | |
| 1 | Identify and analyze problems based on societal and research needs, and evaluate the impact of solutions in societal and environmental contexts for sustainable development. | L2, L4, L5 |
| 2 | Apply knowledge, skills, and standard engineering practices to develop effective solutions to real-world problems. | L3, L6 |
| 3 | Develop interpersonal and teamwork skills, and function effectively as a team member or leader in group activities. | L2, L3 |
| 4 | Draw meaningful inferences from theoretical, experimental, or simulation-based results. | L4 |
| 5 | Demonstrate effective written and oral communication skills in presenting ideas and project outcomes. | L1, L2, L3 |
| 6 | Exhibit self-learning abilities and apply project management principles, promoting continuous and lifelong learning. | L2, L3, L6 |

**Bloom's Taxonomy Levels:**

**1 = Remembering, 2= Understanding, 3 = Applying, 4 = Analyzing, 5 = Evaluating, 6 = Creating**

## LO-PO-PSO MAPPING:

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LO1 | 2 | 3 | 2 | 2 | - | 3 | - | - | - | - | 2 | 3 | 2 |
| LO2 | 3 | 2 | 3 | - | 2 | 2 | - | - | - | - | - | 3 | 3 |
| LO3 | - | - | - | - | - | - | 2 | 3 | 2 | 2 | - | 2 | 2 |
| LO4 | 2 | 3 | - | 3 | 2 | - | - | 2 | 2 | 3 | 3 | 3 | 2 |
| LO5 | - | - | - | - | - | - | - | 2 | 3 | 2 | 2 | 2 | 2 |
| LO6 | - | - | - | - | - | - | - | 2 | 3 | 3 | 3 | 2 | 3 |

## Project topic: PO/PSO Mapping

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Project Topic** | | | | | | | | | | | | | |

# Chapter 1
# Introduction

## 1.1 Introduction:

The modern business landscape has undergone a significant transformation with the rise of data-driven decision-making, where organizations rely heavily on Key Performance Indicators (KPIs) to evaluate performance and guide strategic actions. KPIs such as revenue growth, profit margin, customer churn, and return on investment provide measurable insights into business operations. However, despite these advancements, many organizations still analyze KPIs in isolation or rely on basic visualization tools, which often leads to misinterpretation and poor decision-making. This results in incorrect strategies, financial losses, and inefficient resource allocation. Therefore, there is a growing need for an intelligent system that can analyze KPIs in a more contextual and comprehensive manner.

In recent years, machine learning (ML) and data analytics techniques have shown great potential in improving business intelligence systems. By automatically analyzing KPI data across multiple dimensions, these technologies can identify hidden patterns, detect anomalies, and uncover relationships between different metrics. This project introduces "Metric Lies – KPI Misinterpretation Detection System," an intelligent analytics platform designed to detect misleading KPI patterns and provide meaningful insights for better decision-making.

The system processes structured KPI data from sources such as Excel files and databases, followed by data preprocessing steps including cleaning, normalization, and feature extraction. Key metrics such as revenue growth, profit margin, customer churn rate, customer acquisition cost (CAC), customer lifetime value (CLV), return on investment (ROI), and cash flow are analyzed. These KPIs are further categorized based on company size—large-cap, mid-cap, and small-cap—to enable more context-aware analysis.

Machine learning techniques such as anomaly detection, correlation analysis, and classification are applied to identify unusual trends, conflicting KPIs, and misleading performance indicators. The results are then visualized using interactive dashboards and Power BI, allowing users to clearly interpret KPI behavior and make informed decisions. By combining analytics, machine learning, and visualization, the system enhances the accuracy and effectiveness of business analysis.

The KPI system is a business analytics dashboard that 13nalyses company KPIs to detect anomalies and misleading metrics using rule-based and machine learning techniques

## 1.2 Motivation:

In today's data-driven business environment, organizations rely heavily on Key Performance Indicators (KPIs) to evaluate performance and guide strategic decisions. However, with the increasing volume and complexity of business data, analysts and decision-makers often face challenges in correctly interpreting these metrics. Many times, KPIs are analyzed individually without considering their relationships with other indicators, leading to misleading conclusions. This results in poor decision-making, inefficient resource allocation, and potential financial losses.

While modern analytics tools and dashboards provide visual representations of KPIs, they often lack the ability to detect deeper issues such as conflicting metrics, hidden patterns, or false correlations. For example, a company may observe increasing revenue while profit margins decline, or high customer acquisition with equally high churn rates. Traditional systems fail to highlight these contradictions effectively, making it difficult for analysts to identify the true performance of the business.

The motivation behind the Metric Lies – KPI Misinterpretation Detection System is to develop an intelligent, data-driven solution that can automatically analyze KPIs and detect misleading patterns. By leveraging machine learning techniques such as anomaly detection, correlation analysis, and classification, the system aims to identify inconsistencies and provide meaningful insights into business performance.

The system processes structured KPI data and evaluates key metrics such as revenue growth, profit margin, customer churn rate, customer acquisition cost (CAC), customer lifetime value (CLV), return on investment (ROI), and cash flow. It also considers the context of company size—large cap, mid-cap, and small-cap—to ensure more accurate and relevant analysis.

The ultimate goal of this project is to assist business analysts and decision-makers in understanding the true story behind their data. By highlighting misleading KPIs and providing clear, actionable insights through dashboards and Power BI visualizations, the system enhances decision-making accuracy, reduces risks, and promotes more effective, data-driven strategies in modern organizations.

## 1.3 Problem Statement:

The modern business environment generates a vast amount of data, and organizations rely heavily on Key Performance Indicators (KPIs) to evaluate performance and make strategic decisions. However, with the increasing complexity of business operations, analysts often struggle to interpret KPIs accurately. Traditional analytics systems primarily focus on individual metrics or basic visualizations, which fail to capture the relationships and dependencies between different KPIs. As a result, decision-makers may draw incorrect conclusions from seemingly positive indicators.

For instance, a company may observe an increase in revenue while profit margins decline, or experience rapid customer acquisition alongside high churn rates. Such conflicting KPI patterns are often overlooked in conventional systems, leading to misleading insights and poor business decisions. Additionally, different types of companies—such as large-cap, midcap, and small-cap—have varying priorities, making it even more challenging to interpret KPIs correctly without contextual analysis.

The fundamental problem lies in the lack of an intelligent system that can analyze KPIs collectively, detect anomalies, and identify misleading or conflicting patterns. Existing tools do not provide automated insights into whether a KPI truly reflects business performance or presents a false picture.

To address this issue, the proposed system Metric Lies – KPI Misinterpretation Detection System leverages machine learning and data analytics techniques to process KPI data from structured sources such as Excel files. The system performs anomaly detection, correlation analysis, and classification to identify misleading metrics and highlight inconsistencies. It then presents the results through interactive dashboards and Power BI visualizations, enabling users to better understand business performance and make accurate, data-driven decisions.

## 1.4 Objectives:

- To design and develop an AI-based KPI analysis system capable of detecting misleading patterns and misinterpretations in business metrics.
- To implement data ingestion from structured sources such as Excel files and databases for processing KPI data.
- To apply machine learning techniques such as anomaly detection (Isolation Forest), correlation analysis, and classification models to identify unusual trends and conflicting KPIs.
- To analyze key business metrics including revenue growth, profit margin, customer churn rate, customer acquisition cost (CAC), customer lifetime value (CLV), return on investment (ROI), and cash flow.
- To incorporate company segmentation (large-cap, mid-cap, small-cap) for context-aware KPI analysis.
- To integrate Power BI dashboards for interactive visualization of KPI trends, anomalies, and relationships.
- To develop a user-friendly web interface (frontend) for uploading data and viewing insights without changing the existing UI design.
- To automate insight generation and alert users about misleading or risky KPI patterns.
- To improve decision-making accuracy by providing clear, data-driven insights and reducing dependency on manual analysis.

# Chapter 2

# Literature Survey

## 2.1. Survey of Existing System:

Existing systems for business performance analysis, such as traditional BI tools and dashboards, primarily focus on visualizing KPIs like revenue, profit, and growth. While they help in monitoring performance, they lack intelligent analysis capabilities such as detecting misleading KPI relationships, identifying anomalies, and predicting future trends. Most systems depend on manual interpretation and do not integrate machine learning techniques for automated insights. Additionally, they offer limited customization and do not provide user-specific settings or real-time adaptive analysis. Hence, there is a need for a more advanced system that combines visualization, anomaly detection, and KPI prediction in a single platform.

## 2.2. Limitation Existing system or research gap:

Despite advancements in business analytics and AI-based tools, most existing systems for KPI analysis exhibit several key limitations:

- Lack of Contextual Analysis: Most tools analyze KPIs individually and fail to understand relationships between metrics, leading to incomplete or misleading insights.
- No Detection of KPI Conflicts: Existing systems cannot identify contradictory patterns such as high revenue growth with low profitability or high customer acquisition with high churn rates.
- Over-Reliance on Visualization: Dashboards like Power BI and Tableau focus on displaying data rather than interpreting it, requiring manual effort from analysts to derive conclusions.
- Limited Use of AI for Interpretation: While some tools use machine learning for prediction, they do not focus on detecting misleading KPIs, false correlations, or anomaly patterns.
- Lack of Business Context Awareness: Current systems do not differentiate between large-cap, mid-cap, and small-cap companies, resulting in generalized insights that may not be accurate for all business types.

## 2.3. Mini Project Contribution:

### Silas Shelly

Developed core machine learning logic for KPI analysis, including anomaly detection, correlation analysis, and classification of misleading metrics.

### Shannel D'Costa

Integrated the frontend with backend systems and ensured smooth data flow between user input and analysis modules.

### Aidan Rebello

Built data preprocessing modules, including Excel data handling, cleaning, normalization, and KPI feature extraction.

### Aryan Gupta

Designed and developed the frontend interface, including dashboard layout, user input pages, and overall UI/UX.

# Chapter 3
# Proposed System

## 3.1. Architecture/ Framework:

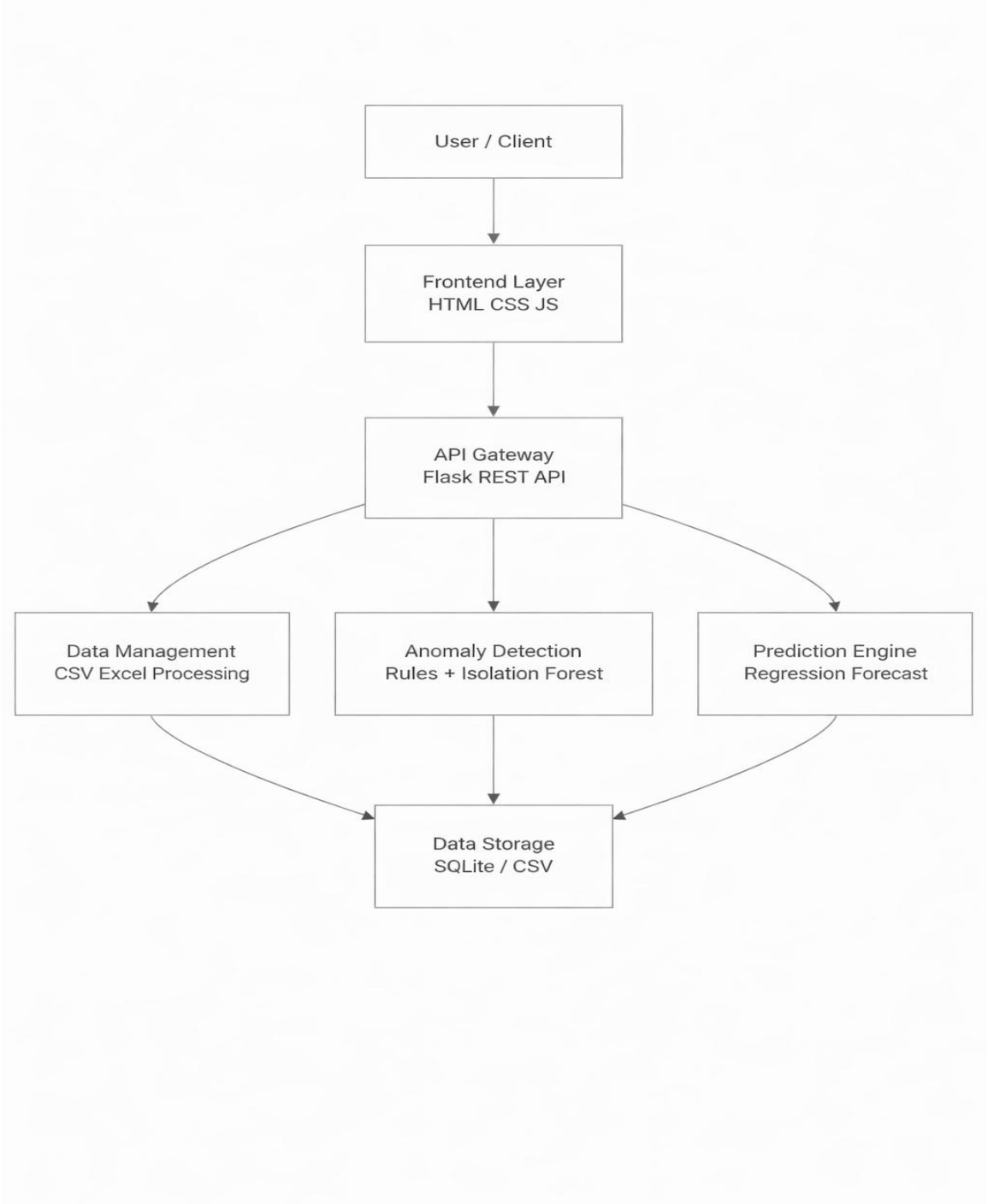


Fig 3.1: Architecture/Framework

## 3.2. Algorithm and Process Design:

1. Start
2. Load company KPI dataset from Excel/CSV or database.
3. Preprocess data (handle missing values, normalize if required).
4. For each company in dataset:
   - Read KPI values (revenue_growth, profit_margin, churn_rate, CAC, CLV, ROI, cash_flow)
5. Apply rule-based checks:
   - If revenue_growth ↑ and cash_flow ↓ → flag anomaly
   - If CLV < CAC → flag anomaly
   - If churn_rate high → flag anomaly
6. Apply machine learning model (Isolation Forest):
   - Input KPI values
   - Detect outliers/anomalies
7. Assign risk score (Low / Medium / High) based on anomalies.
8. Store results.
9. Apply prediction model:
   - Use regression or trend analysis.
   - Predict future KPI values.
10. Send processed data to frontend dashboard.
11. Display results (charts, tables, alerts).
12. Stop

**Process Workflow:**

1. **Data Input Process:**
   • User uploads Excel/CSV file or system loads predefined data • Data includes company KPIs

2. **Data Processing Process:**
   • Clean and validate data
   • Convert into structured format using pandas

3. **Anomaly Detection Process:**
   • Apply rule-based logic
   • Apply ML model (Isolation Forest)
   • Identify misleading KPI relationships

4. **Prediction Process:**
   • Use regression or simple trend formulas
   • Forecast future KPI values 19

5. **Data Storage Process:**
   • Store data in CSV/SQLite
   • Maintain processed results

6. **Visualization Process:** • Send data via API to frontend
   - Display:
   - KPI dashboard
   - Charts
   - Anomaly alerts

## 3.3. Details of Hardware & Software:

**Hardware Requirements:**

- **Processor:** Intel i3 or above
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** Minimum 500 MB free space
- **Display:** 1366 × 768 resolution or higher
- **Network:** Internet connection (for libraries and updates)

**Software Requirements:**

- **Operating System:**
  Windows 10/11 or Linux (Ubuntu)

- **Programming Language:**
  Python (for backend processing and ML)

- **Backend Framework:**
  Flask (for API development)

- **Frontend Technologies:**
  HTML, CSS, JavaScript
  Chart.js (for data visualization)

- **Libraries & Tools:**
  pandas (data processing)
  scikit-learn (machine learning – Isolation Forest)
  openpyxl (Excel file handling)

- **Database:**
  SQLite / CSV (for storing KPI data)

- **Development Tools:** VS Code (code editor)
  Web Browser (Chrome/Edge)

## 3.4. PERT / GANTT / CPM Charts:

### PERT Chart:

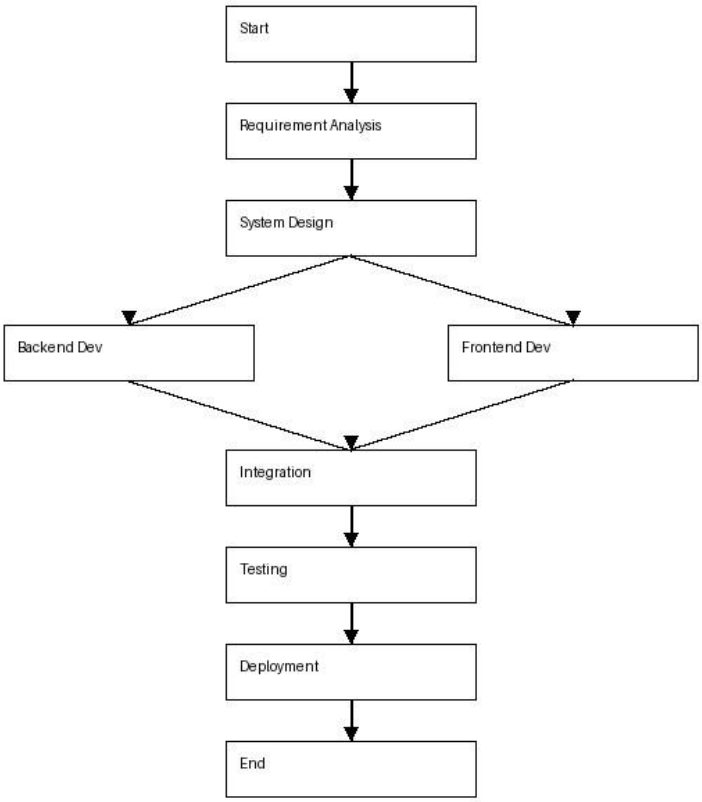


Fig 3.2: PERT Chart

**GANTT Chart:**

| Task | Duration | Week 1 | Week 2 | Week 3 | Week 4 |
|---|---|---|---|---|---|
| Requirement Analysis | 3 days | | | | |
| System Design | 4 days | | | | |
| Backend Development | 6 days | | | | |
| Frontend Development | 6 days | | | | |
| Integration | 3 days | | | | |
| Testing & Debugging | 4 days | | | | |
| Final Deployment & Report | 4 days | | | | |

Fig 3.3: Gantt Chart

**CPM Charts:**

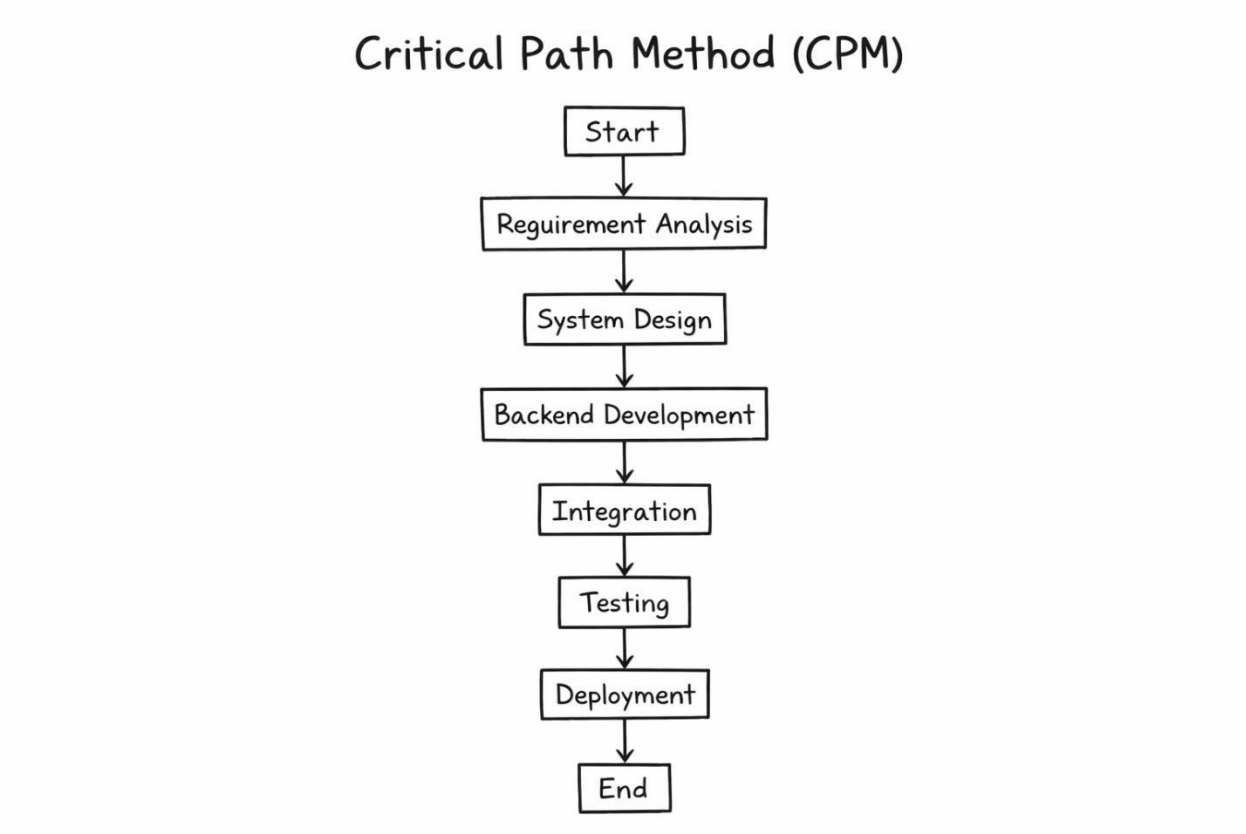


Fig 3.4: CPM Chart

# Chapter 4
# Results and Discussions

## 4.1 Results:

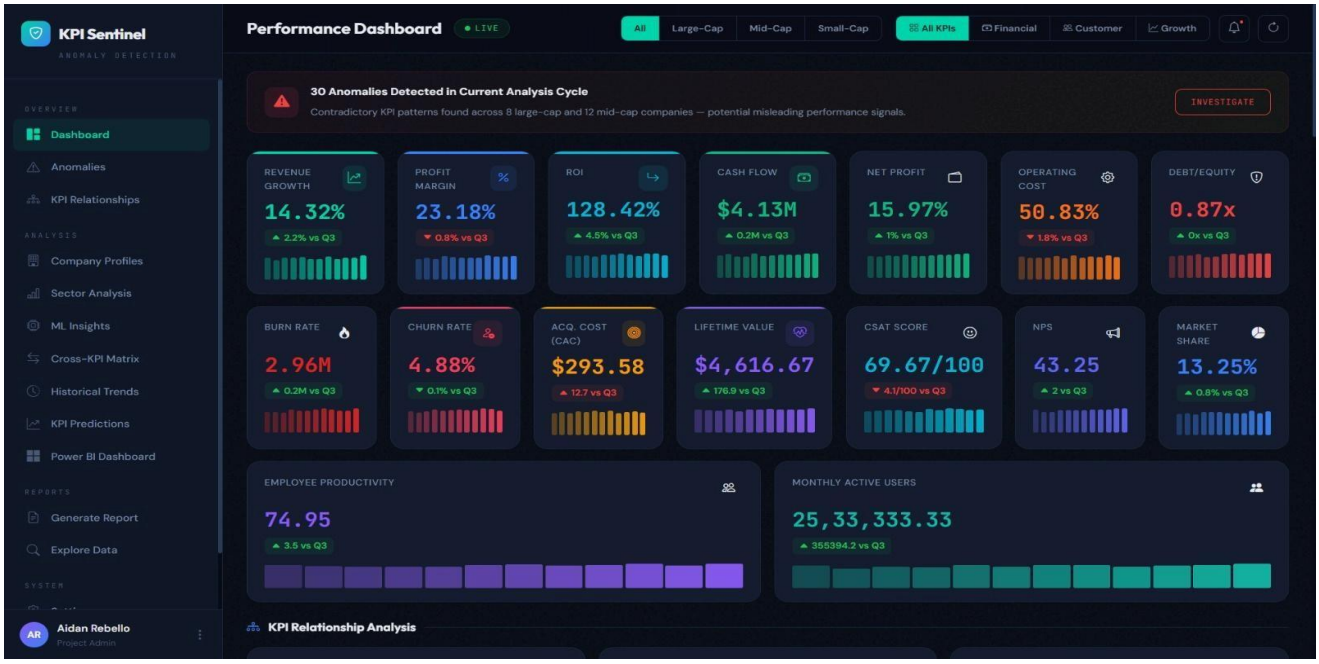


Fig 4.1: KPI Sentinel dashboard

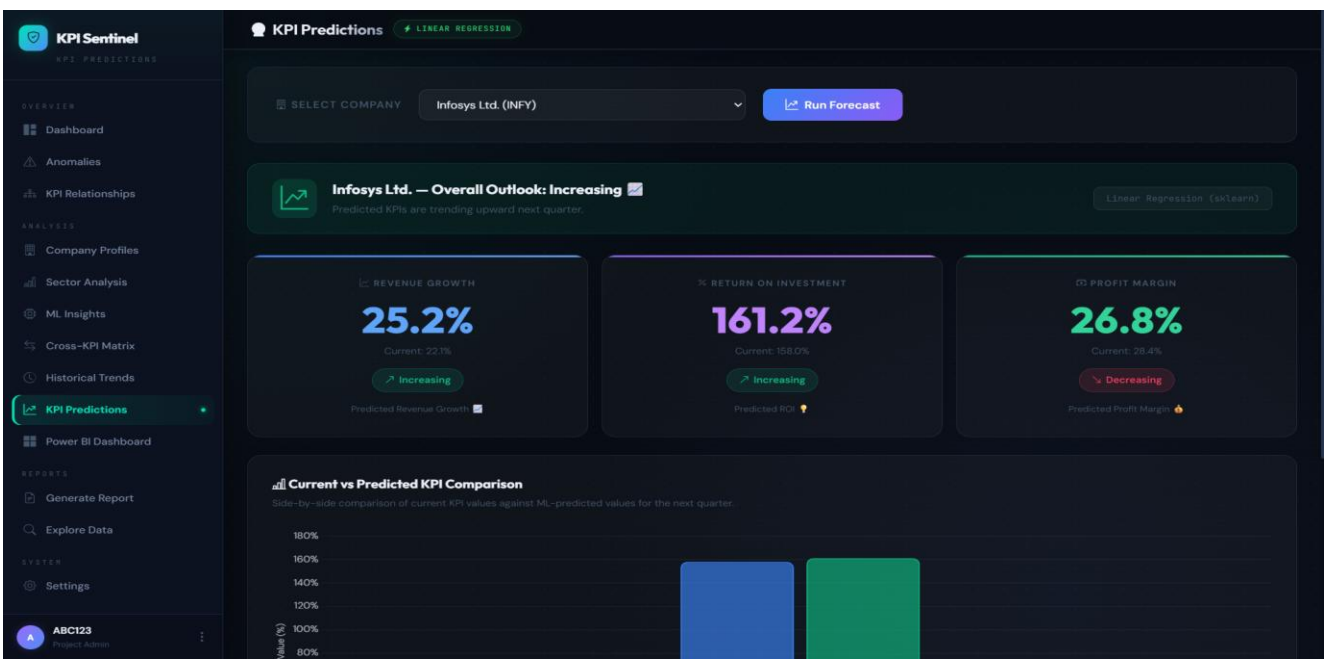


Fig 4.2: KPI Predictions

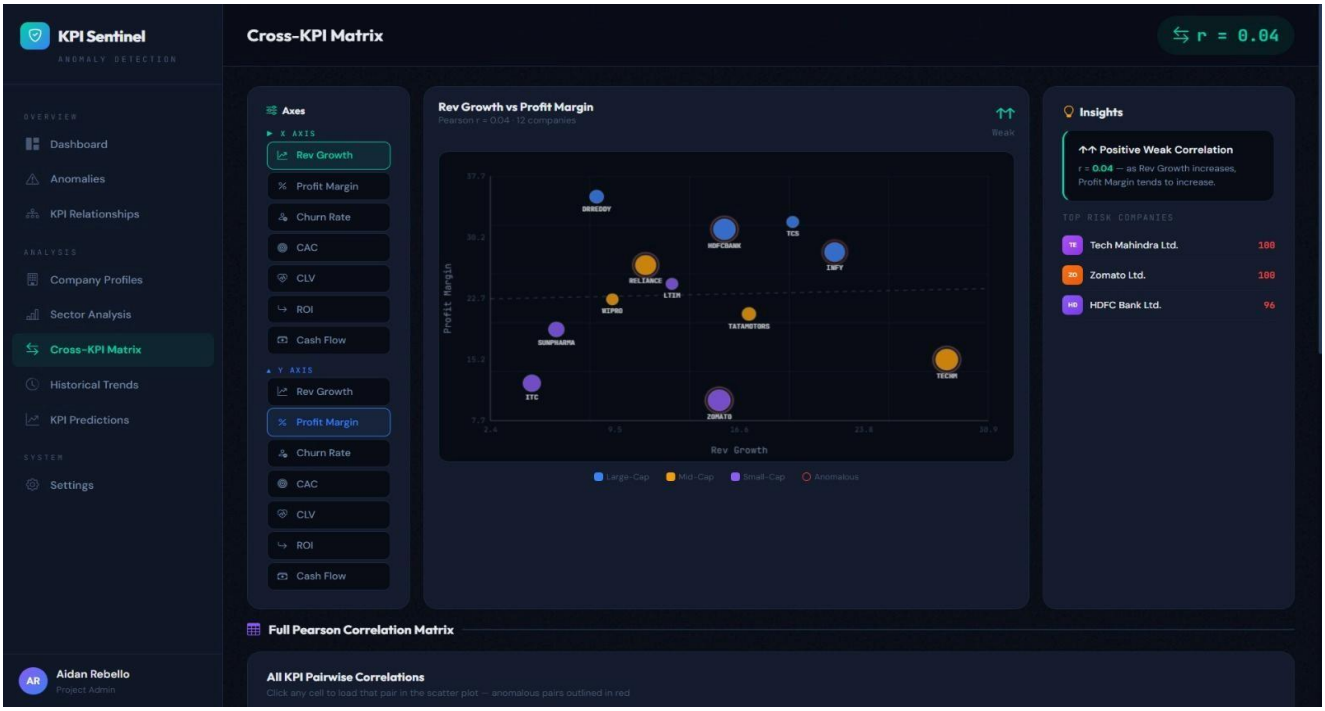

Fig 4.3: Cross-KPI Matrix

## 4.2 Discussions:

- **Company KPI data was loaded and processed successfully.**
- **Rule-based and machine learning methods were used to detect anomalies.**
- **System correctly identified misleading KPIs and assigned risk levels.**
- **KPI prediction module generated future trends.**
- **Results were displayed using an interactive dashboard with charts and tables.**

**Performance Metrics:**

- **Accuracy:** Measures how correctly anomalies are detected.
- **Precision:** Checks how many detected anomalies are actually correct.
- **Recall:** Measures how many real anomalies are identified.
- **F1-Score:** Balances precision and recall.
- **MAE (Mean Absolute Error):** Measures prediction error of KPIs.
- **Response Time:** Time taken to process data and show results.

# Chapter 5

# Conclusion and Future Scope

## Conclusion:

The KPI Sentinel system successfully analyzes company performance using multiple KPIs and detects misleading patterns through rule-based and machine learning techniques. It provides risk assessment, anomaly detection, and KPI prediction in an interactive dashboard, helping users make better business decisions. The system is efficient, user-friendly, and suitable for real-time business analytics. The KPI Sentinel system provides an effective solution for analyzing company performance by evaluating multiple KPIs together instead of in isolation. It successfully identifies misleading patterns using rule-based logic and machine learning techniques such as Isolation Forest. The system also includes a prediction module that forecasts future KPI trends, helping in proactive decision-making. With its interactive dashboard and clear visualization of results, the system enhances understanding of business performance and supports better financial analysis. Overall, it is a reliable, efficient, and user-friendly tool for detecting KPI misinterpretation and improving business insights.

## Future Scope:

The KPI Sentinel system can be further enhanced by integrating real-time data sources such as financial APIs to provide live company insights. Advanced machine learning models can be implemented to improve prediction accuracy and enable deeper trend analysis. The system can also be expanded with features like user role management, cloud deployment, and mobile application support for wider accessibility. Additionally, incorporating AI-based automated insights and natural language summaries can help users better understand complex KPI relationships and make faster business decisions.

# References

[1]. Breunig, M. M., Kriegel, H. P., Ng, R. T., & Sander, J. (2000). *LOF: Identifying density-based local outliers*. ACM SIGMOD.

[2]. Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008). *Isolation Forest*. IEEE International Conference on Data Mining.

[3]. Han, J., Kamber, M., & Pei, J. (2011). *Data Mining: Concepts and Techniques* (3rd ed.). Morgan Kaufmann.

[4]. scikit-learn Developers. (2024). *Scikit-learn: Machine Learning in Python*. Retrieved from https://scikit-learn.org

[5]. Flask Documentation. (2024). *Flask Web Development Framework*. Retrieved from https://flask.palletsprojects.com

[6]. pandas Documentation. (2024). *Pandas: Data Analysis Library*. Retrieved from https://pandas.pydata.org

[7]. Chart.js Documentation. (2024). *Chart.js Visualization Library*. Retrieved from https://www.chartjs.org

[8]. Chart.js Documentation. (2024). *Chart.js Visualization Library*. Retrieved from https://www.chartjs.org